

TRACE-ECG: Transfer-Based Re-Identification Attack on Cross-Dataset ECG Signals

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Abstract. Health data are increasingly shared for clinical research and through the interconnection of healthcare systems. At the same time, they are arguably the most sensitive form of personal information, embedding rich clinical and biometric attributes. Although de-identification and pseudo-anonymization are applied prior to release, empirical evidence suggests that re-identification risks persist. Electrocardiograms (ECGs), in particular, contain stable temporal and morphological features. Prior ECG re-identification studies evaluate models trained and tested within the same cohorts, implicitly assuming the adversary has access to the target population data during training. In this work, we remove this assumption by proposing a strict cross-dataset re-identification attack where the adversary uses a transfer model on an external dataset. We evaluate the transferability of identity representations learned using an encoder trained exclusively on one population to generalize to unseen cohorts and subsequently freeze it before deployment on fully disjoint external datasets, including ECG-ID, MIMIC-IV-ECG subset, MIT-BIH, and PTB-XL. To assess robustness, we benchmark two configurations (MIT-BIH and MIMIC-IV-ECG as sources) without target supervision. Under this realistic adversarial setting, we perform identity linkage using deep neural embeddings and lead-aware similarity scoring. Across both source configurations, multi-lead fusion achieves consistently high Top-1 accuracies (≈ 85 -96%) on external cohorts, while single-lead performance varies (≈ 56 -100%). Our findings demonstrate that ECG identity structure remains exploitable even in the absence of identity overlap, target supervision, or data adaptation, highlighting a generalizable privacy risk and matching results reported in within-dataset evaluations. Beyond quantifying attack effectiveness, we analyze amplification effects introduced by multi-lead aggregation and frame de-anonymization as a measurable risk assessment problem. We structure an aligned roadmap linking attack testing to Privacy Enhancing Technologies (PETs).

Keywords: Re-identification Attack · De-anonymization · Healthcare Privacy · Electrocardiogram (ECG)

1 Introduction

Digital health systems increasingly rely on patient data to enable clinical research, Artificial Intelligence (AI) development, and connected infrastructures.

Among these data types, electrocardiograms (ECGs) are widely used due to their value in cardiology and patient monitoring. ECG repositories are frequently released in de-identified form under the assumption that removing explicit identifiers is sufficient to protect patient privacy. However, this does not remove the embedded biometric structure within the signal. ECGs encode stable temporal and morphological patterns (e.g., PQRST morphology), motivating studies of authentication and identity verification systems [23,39]. At the same time, they introduce privacy risks related to identity attacks [16,3]. Under regulatory frameworks such as the General Data Protection Regulation (GDPR) [13], anonymized data must not be re-identifiable by any means. However, the rapid advancement of Deep Learning (DL) has significantly raised the de-anonymization threat. Prior studies have demonstrated high identification performance within individual ECG datasets [14]. However, these evaluations rely on intra-dataset splits or authentication-style [8] validation in which models are trained within the same population to perform re-identification. This confirms the presence of exploitable information, yet it does not capture a realistic cross-institutional adversary. This gap motivates two research questions we cover in our work: Does ECG identity information persist across independent populations when exploiting external data? Can a frozen encoder enable record-to-identity linkage across datasets without target supervision?

To answer these questions, we design a strict cross-dataset re-identification protocol. In practice, an attacker trains an encoder on a single subset source cohort and subsequently frozen, then deploys it unchanged on external datasets [17], including ECG-ID [31], MIMIC-IV [18,41], MIT-BIH [36], and PTB-XL [44], with explicit source/target separation. We evaluated our pipeline using MIT-BIH and MIMIC-IV as sources, and we observed high linkage performance in both settings. In multi-lead datasets, fusion reports Top-1 accuracies ranging from 85% to 96% on external populations. Our findings demonstrate that ECG identity representations generalize across external datasets to remain exploitable under realistic adversarial assumptions, without target supervision or dataset-specific adaptation. Despite the more restrictive assumptions imposed by the attacker, we retain high levels of re-identification comparable to existing, less-constrained within-cohort matching attacks in prior work.

We further outline a mitigation roadmap that connects adversarial evaluation with privacy-preserving mechanisms. Our contributions include:

- Strict frozen-transfer re-identification threat model with explicit source-target identity disjointness and no target supervision.
- We evaluate two independent source configurations (MIT-BIH and MIMIC-IV) and test them on the four datasets to assess robustness to source selection and to benchmark the attack across datasets.
- Analysis single-lead and multi-lead linkage evaluation and leads fusion accuracy amplification.
- We introduce post-hoc confidence analysis on the linkage scores, providing a bridge toward open-world ECG linkage.
- Risk framing and mitigation roadmap: We frame ECG re-identification as a measurable privacy risk and outline an attack-aligned mitigation roadmap.

The remainder of the paper is organized as follows. Section 2 reviews related work on re-identification and ECG biometrics. We then describe datasets, threat model, and attack methodology in Section 3. Section 4 presents our linkage performance and analysis and Section 5 outlines mitigation and privacy-by-design considerations, followed by conclusions in Section 6.

2 Background and Related Work

This section highlights prior work on de-anonymization in health data, ECG-based identification, and privacy risks. We focus particularly on threat models and evaluation protocols relevant to signal-matching re-identification.

2.1 De-Anonymization in Health Data

De-anonymization refers to linking, inferring information, or reverse anonymization to re-identify individuals or link their records. Early work by Sweeney [43] demonstrated how information inference can break anonymization even when explicit identifiers are removed. Since then, de-anonymization studies have expanded across domains, including social networks [7], mobility traces [15], as well as healthcare in related surveys [4,22]. In healthcare, re-identification risks have evolved substantially. Early empirical works reported moderate success [12]. In contrast, more recent studies show greater vulnerabilities across health data types (e.g., ECG, Genomics) and an increase in empirical studies [2] demonstrating that DL and data availability have increased the feasibility of attacks. However, studies focus on structured data, and the physiological signals (Vital Signs data or ECGs) remain underexplored from a strict cross-dataset threat perspective.

2.2 ECG as a Biometric Signal and Re-Identification Risk

ECG signals are widely used in cardiology for monitoring and diagnosing arrhythmias and other cardiac conditions, but beyond utility, they contain stable morphological patterns (e.g., QRS complex, RR intervals) that disclose inter-individual variability [14]. This property has motivated research in ECG-based authentication [10,35] and studies have reported accuracies often exceeding 90% and reaching 100% for some cases [5,42,6]. Several works have also proposed full DL frameworks for ECG biometric recognition. For instance, [34] combines CNN architectures to perform verification across ECG databases. However, these studies use intra-dataset with overlapping records, to confirm the presence of a biometric structure rather than de-anonymization. More recent work investigates re-identification in ECG data, such as [16], which uses a DL model; [45], which studies interpretability and linkage risk across ECG datasets; and [46], which demonstrates the feasibility of re-identification using a linkage attack. We extend the literature by proposing a stricter cross-dataset transfer scenario, where

the encoder is trained on a source population and deployed unchanged on external data. To our knowledge, no prior work has evaluated ECG re-identification in a frozen cross-dataset transfer setting with explicit source–target identity disjointness.

2.3 De-Anonymization Threat Models vs Authentication

It is important to distinguish biometric authentication from de-anonymization. Authentication systems aim to verify or identify enrolled users in a controlled, cooperative setting, using training and evaluation data from the same population. In contrast, de-anonymization attacks attempt to link de-identified records without subject cooperation. Such systems aim to authenticate enrolled users within controlled settings where training and evaluation occur on the same populations, a process that differs fundamentally from cross-dataset linkage. Consequently, realistic re-identification evaluations require strict separation between source and target populations and explicit identity disjointness to avoid bias.

2.4 Privacy Regulation and Risk Assessment

When assessing re-identification and identifiability as a privacy mechanism, the GDPR refers to the ‘means reasonably likely to be used..’ for identification [13]. Although it lacks a straightforward guide, we imply that anonymization must be evaluated against reasonable capabilities rather than theoretical assumptions. Classical privacy research has demonstrated that even limited query access may reveal sensitive information if release mechanisms are not carefully designed [11]. Furthermore, traditional anonymization can fail when faced against linkage attacks that exploit quasi-identifiers or inferred information [47], resulting in non-compliance with regulations, particularly GDPR. We emphasize the necessity of realistic risk evaluation for de-identification, rather than theoretical assumptions, as DL enables high-capacity models to extract latent biometric structure from physiological signals such as ECG.

3 Threat Model and Re-identification Attack

This section formalizes the re-identification protocol used in our study. We define the patient-level linkage setting, describe the dataset split construction, adversary model, assumptions and constraints, and then detail the attack pipeline and scoring protocol. The objective is to evaluate whether the leaked ECG queries can be linked to de-identified records under explicit source–target identity disjointness and without target fine-tuning or target-label supervision. At the linkage level, we refer to the attacker-identified records as *leaked records* (also called leaked queries), corresponding to ECGs whose patient identities are known to the adversary, and the de-identified records as the *reference set*.

3.1 Attack Overview

We consider a patient-level linkage setting in which each identity u corresponds to a patient identifier (a patient may have one or multiple records). The attacker’s goal is to train a general attack model (on any openly available dataset) that is able to generate unique patient fingerprints, and then use this attack model to re-identify patients on the target dataset, a small part of which has been leaked. As an overview, the attack proceeds in three stages. First, an encoder is trained on an external data source to learn discriminative ECG representations. Second, the encoder is frozen and applied to the target dataset to construct patient-level fingerprints. Third, each query fingerprint is matched against the target reference fingerprints, and the candidate identities are ranked.

3.2 Datasets and Split Construction

We evaluate our attack on four public ECG datasets with diverse settings: ECG-ID [30], MIMIC-IV-ECG subset [18], MIT-BIH [36], and PTB-XL [44]. ECG-ID provides 20-second single-lead recordings for 90 individuals, with multiple sessions per subject (up to 20) over several months. MIMIC-IV-ECG includes 12-lead recordings collected across different hospital visits, spanning days and months. MIT-BIH consists of 30-minute recordings acquired using a dual-lead setup for 47 patients. PTB-XL provides large-scale 12-lead recordings with over 1000 patients, representing a high-variability cohort.

Across datasets, identity is defined at the patient level. The diverse sets expose heterogeneous conditions, including single versus multi-lead acquisition, short versus long recordings, and temporal sparsity spanning over days, months, and in some cases years, in addition to the clinical composition. ECG-ID is a biometric-style dataset, MIT-BIH is arrhythmia-enriched, with recordings including common and less common clinical arrhythmias, MIMIC-IV-ECG contains diagnostic ECGs from hospital patients and PTB-XL includes both normal ECGs and multiple diagnostic classes. Therefore, pathology may contribute to identity separability. However, even with the cohorts’ different acquisition settings, our results cannot be explained by memorization of pathology distribution. We interpret the results as evidence of transferable ECG identity structure under heterogeneous conditions. To simulate the adversarial setting, each target dataset is partitioned into a de-identified set and an identified query set. We construct three deterministic splits per dataset. For datasets with multiple records per patient, disjoint subsets of records are assigned to the leaked and reference roles, following an approximate 25%/75% split when sufficient records are available. The exact proportion is dataset-dependent and determined at the patient level. Assignments are rotated across three splits, and patients with insufficient records are excluded. MIT-BIH does not provide multiple independent records per patient. To approximate our setting, each long recording is segmented into seven spaced sub-records, of which five are assigned to the reference side and two to the leaked side. Because these sub-records originate from the same continuous recording, the leaked and reference sub-records might potentially have a higher

level of inner similarity, compared to other datasets in the study. However, experimental evidence confirms that this does not imply a higher re-identification rate, as we achieve comparable results using datasets for which records are obtained from different sessions (such as ECG-ID and PTB-XL fused). We also include the MIT-BIH dataset because it is a well-established benchmark in the relevant literature [25,3,46,28].

Signal preprocessing and snippet extraction: We apply standard preprocessing techniques from the established ECG literature. ECG signals are filtered with a bandpass Butterworth filter to reduce baseline wander, powerline interference, and motion artefacts while preserving the main waveform morphology for representation learning [35,34]. R-peaks are detected using the PanTompkins method [38] as implemented in NeuroKit2 [32]. Around each detected R-peak, a fixed-length heartbeat snippet is extracted using a centred window of 256 samples, following the common ECG beat-level representation strategy used in heartbeat classification and transferable ECG representation learning [9,26,24].

3.3 Threat Model

We consider a transfer-based threat model in which the adversary trains an ECG encoder on a source dataset population and deploys it unchanged on unseen target populations from external datasets. The objective is to link ECG traces belonging to the same patient within an external dataset, even with sparse sampling, temporal separation (months to years), and independent acquisition conditions. To assess our attack’s robustness, we instantiate this protocol twice: (i) MIT-BIH split1 as source, and (ii) MIMIC-IV split1 as source.

Let $\mathcal{D}_S$ represent the source dataset and $\mathcal{D}_T^{(k)}$ the external target datasets. When MIT-BIH is used as the source, the external targets are ECG-ID, MIMIC-IV, and PTB-XL; when MIMIC-IV is the source, the external targets are ECG-ID, MIT-BIH, and PTB-XL. Same-dataset held-out splits evaluations are reported as secondary stability analysis as detailed in Section 4 to highlight that the fingerprint construction remains stable under disjoint splits from the same dataset family. Identity sets are denoted by $\mathcal{I}_S$ and $\mathcal{I}_T^{(k)}$, with:

$$\mathcal{I}_S \cap \mathcal{I}_T^{(k)} = \emptyset, \quad \forall k.$$

- **The adversary has access to:**
 1. A leaked (identified) source dataset for training the encoder (a compromised third party).
 2. De-identified external targets (published datasets).
 3. Leaked queries (compromised/identified records).
- **The adversary’s goal:** For each leaked query, rank candidate reference identities, and recover the correct patient match at top- k .
- **Realistic scenario:** A hospital releases de-identified ECG records for a research study (includes other sensitive information). Separately, an attacker obtains identified ECGs via compromised devices or a secondary data flow.

Even without overlapping patients between source and target datasets, the attacker trains a biometric encoder and attempts to link queries to reference targets.

The proposed setting is deliberately more challenging and realistic than within-cohort ECG re-identification as in prior work that trains and evaluates models within the same dataset or allows adaptation to the target population [16,45,3]. Our attack requires identity representations learned on a source population to generalize to fully disjoint external datasets with different cohort and acquisition characteristics. This is enforced by (i) complete identity disjointness between source and target datasets, (ii) no access to target identity labels during training, and (iii) no fine-tuning to the target distribution. As a result, successful linkage cannot be attributed to dataset memorization or target-specific optimization, but instead reflects the transferability of identity-related ECG structure.

3.4 Adversary Knowledge and Constraints

The adversary is assumed to have general knowledge of ECG acquisition information (e.g., signal format, sampling frequency) and of signal morphology (e.g., PQRST structures). The attacker also knows which records constitute the public reference records versus newly observed, identified queries with access to target ECG signals only (no demographic or auxiliary metadata). The adversary does not have:

- Target identity labels for model training.
- Source/target identity overlap.
- Target-specific encoder fine-tuning.

These constraints ensure that the attack is evaluated under cross-dataset transfer, without access to target identity labels, overlap, or patient adaptation.

3.5 Attack Methodology

The overall pipeline is illustrated in Figure 1. We trained a 1D ResNet identity encoder [21] on the source only using categorical cross-entropy over source identities as summarized on Figure 2. The embedding layer (dimension = 256) is then used as the biometric representation. The CNN (Convolutional Neural Network) encoder operates on a single channel, each snippet has a shape $\mathbb{R}^{256 \times 1}$. The encoder parameters are then frozen and reused without target fine-tuning, retraining, or access to target identity labels. Heartbeat snippets from the source records are divided into an 80/20 train/validation split at the snippet level using a fixed random seed of 42. The validation subset is used only to monitor model performance. For multi-lead datasets, the encoder is applied independently to each lead, and lead-wise similarity scores are fused afterwards.

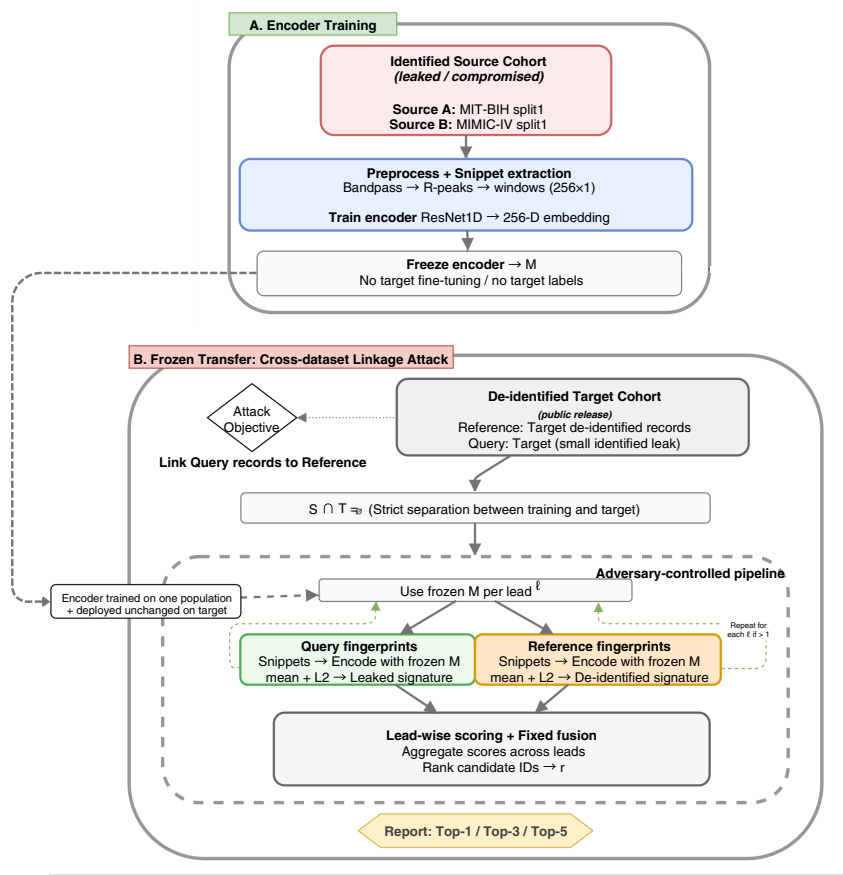


Fig. 1. Frozen Transfer Linkage Attack for Cross-Dataset ECG Re-Identification

1. **Fingerprint construction:** For each identity u and lead ℓ , we filter the ECG signal, detect R-peaks, extract a set of snippets $\{x_{u,\ell,j}\}_{j=1}^{n_{u,\ell}}$ and encode them using the frozen encoder f_θ . Embeddings are averaged and L2-normalized to produce one lead-specific fingerprint per patient. In the adversarial setting, identities associated with the reference set are treated as unknown to the attacker during matching, and are only used internally to construct reference fingerprints for evaluation:

$$\mathbf{z}_{u,\ell} = \frac{\frac{1}{n_{u,\ell}} \sum_{j=1}^{n_{u,\ell}} f_\theta(x_{u,\ell,j})}{\left\| \frac{1}{n_{u,\ell}} \sum_{j=1}^{n_{u,\ell}} f_\theta(x_{u,\ell,j}) \right\|_2}.$$

Reference fingerprints are constructed from the de-identified split, while query fingerprints are constructed from the leaked queries.

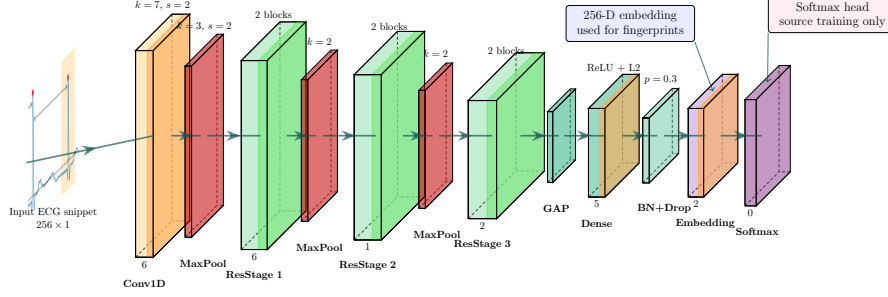


Fig. 2. ECG Attack Encoder. Each stage contains 2 residual blocks. GAP refers to global average pooling, and BN+Drop is batch normalization followed by dropout.

2. **Distance-based matching:** For a query identity q and a reference identity r on lead ℓ , we compute the Euclidean distance between the corresponding patient fingerprints

$$d_{\ell}(q, r) = \|\mathbf{z}_{q,\ell}^Q - \mathbf{z}_{r,\ell}^R\|_2.$$

Candidates are ranked in ascending order of $d_{\ell}(q, r)$, following embedding-space biometric linkage practice [29,33]. To perform a confidence analysis and lead fusion, we convert each distance to a simple bounded score

$$s_{\ell}(q, r) = \frac{1}{1 + d_{\ell}(q, r)}.$$

For multi-lead targets, the final score is obtained by averaging lead-wise scores. This transform does not change the underlying nearest-neighbor matching, it is used only to express match strength and aggregate evidence.

3. **Multi-lead fusion:** We compute a similarity score independently when multiple leads are available (case of MIMIC-IV and PTB-XL), for each lead and then average these scores across all leads. The final similarity between a query and a reference identity is therefore the mean of the lead-wise similarities:

$$S(q, r) = \frac{1}{L} \sum_{\ell=1}^L s_{\ell}(q, r).$$

Reference identities are then ranked in descending order of $S(q, r)$. For single-lead datasets, the lead-specific score is used directly.

4. **Double benchmarking for consistency:** To assess dependence on the source population, we run the full protocol with two sources (MIT-BIH→targets and MIMIC-IV→targets) using an identical pipeline, scoring, and evaluation. This provides a source-consistency check for cross-dataset linkage performance.

Separately from the linkage attack and scoring, we added a confidence analysis layer to map the top-1 score to a more reliability/confidence value. We use:

$$\hat{p}(\text{correct} \mid s_1, m) = \sigma(\alpha s_1 + \beta m + \gamma),$$

where $\sigma(\cdot)$ represents the logistic sigmoid. This calibration layer [19] is used only for descriptive confidence analysis and threshold-based rejection as detailed in Section 4.4 and does not affect the core matcher nor the re-identification results. The calibration model is not trained on a separate held-out calibration set.

The attack pipeline implementation is available through a controlled-access repository.¹

4 Results and Analysis

In this study we follow explicit source-target identity matching. For each dataset, we evaluate three deterministic split variants to assess the consistency of the attack. Performance is reported using Top-1-3-5 under closed-world matching between reference records and query samples. We break down our attack results into single-lead and multi-lead on the four datasets, summarized in Section 4.1. To assess robustness to source selection, we repeated the protocol using MIT-BIH and MIMIC-IV as sources. Both configurations use identical fingerprint construction, scoring, fusion, and evaluation rules.

4.1 Linkage Performance

Transfer using MIMIC-IV $\rightarrow$ Targets: In Table 1, we report performance when training the encoder on MIMIC-IV split1. Single-lead Top-1 accuracy ranges from 66.41% on PTB-XL to 97.87% on MIT-BIH. When fusing all available leads in 12-lead datasets, Top-1 accuracy increases consistently, reaching between 90.56% and 96.10%.

Table 1. Linkage Attack Accuracies (MIMIC-IV as source). T1, T3, and T5 denote Top- k accuracy and their values are in %.

Target Split	Leads	Size	Best Lead	Single T1	Fused T1	Single T3	Single T5
ECG IDB/split1	1	89	I	93.26	N/A	96.63	96.63
ECG IDB/split2	1	89	I	93.26	N/A	96.63	96.63
ECG IDB/split3	1	89	I	94.38	N/A	95.51	96.63
MIT-BIH/split1	1	47	MIT channel-0	95.74	N/A	97.87	100.00
MIT-BIH/split2	1	47	MIT channel-0	97.87	N/A	100.00	100.00
MIT-BIH/split3	1	47	MIT channel-0	97.87	N/A	100.00	100.00
PTB XL/split1	12	521	II	68.91	91.75	95.78	96.55
PTB XL/split2	12	521	I	66.41	90.56	95.10	96.16
PTB XL/split3	12	521	II	69.48	92.51	95.20	96.16
MIMIC-IV/split2	12	77	II	89.61	94.81	94.81	96.10
MIMIC-IV/split3	12	77	I	90.90	96.10	98.70	98.70

Note: Grey rows indicate secondary attack effectiveness checks under disjoint splits from the same dataset cohort, while the primary claim is based on the stricter external-dataset evaluations.

¹ TRACE-ECG Implementation Repository.

Transfer using MIT-BIH $\rightarrow$ Targets: We further train the encoder on MIT-BIH split1 and evaluate transfer to external cohorts under identical frozen conditions, enabling benchmarking comparison with the MIMIC-source configuration. In Table 2, we detail the stable results obtained, with a single-lead best accuracy of up to 91.01% and a max of 89.64% when fusing leads on external targets (not considering other MIT-BIH splits).

Table 2. Linkage Attack Accuracies (MIT-BIH as source). T1, T3, and T5 denote Top- k accuracy and their values are in %.

Target Split	Leads	Size	Best Lead	Single T1	Fused T1	Single T3	Single T5
ECG IDB/split1	1	89	I	86.52	N/A	94.38	97.75
ECG IDB/split2	1	89	I	91.01	N/A	95.51	96.63
ECG IDB/split3	1	89	I	86.52	N/A	94.38	95.51
MIMIC-IV/split1	12	77	II	66.23	89.61	93.51	93.51
MIMIC-IV/split2	12	77	II	72.73	89.61	93.51	93.51
MIMIC-IV/split3	12	77	I	62.34	89.61	93.51	94.81
PTB XL/split1	12	521	II	55.66	89.25	93.09	94.43
PTB XL/split2	12	521	I	56.24	85.03	91.17	93.67
PTB XL/split3	12	521	II	59.88	89.64	93.47	95.20
MIT-BIH/split2	1	47	MIT channel-0	100.00	N/A	100.00	100.00
MIT-BIH/split3	1	47	MIT channel-0	100.00	N/A	100.00	100.00

Note: Grey rows indicate secondary attack effectiveness checks under disjoint splits from the same dataset cohort, while the primary claim is based on the stricter external-dataset evaluations.

4.2 Performance Analysis

Beyond raw accuracy metrics, below we break down the behaviour of our transfer attack, the amplification effects of multi-lead setting, and the implications for privacy risk.

- **Transferability of identity representations:** Across both configurations, the encoder maintains high performance on disjoint patient pools, indicating that the ECG representations capture persistent cross-domain structure.
- **Single-lead vs multi-lead:** In multi-lead targets (MIMIC and PTB-XL), single-lead performance varies depending on the source cohort, whereas lead fusion consistently raises Top-1 accuracy into the 85–96% range across both configurations. The linkage improvement is explained by the complementary identity patterns captured, and aggregation stabilizes the very noisy or weak lead matches.
- **Cross-dataset stability:** Fusion performance is consistent across sets and sources. Over all evaluated datasets, the best Top-1 accuracy on targets ranges from 85.03% to 97.87%, with mean values of 94.13% (MIMIC-source) and 88.53% (MIT-source). The attack remains robust against varying attacks through data splits, for instance, ECG-ID single-lead Top-1 is 94.38% as the higher accuracy and 93.26% as the lowest, and for PTB-XL using fused leads, 92.51% represents the highest and 90.56% as the lowest, both with MIMIC as the source. Similarly, when MIT-BIH is the attack source, ECG-ID’s best

is 91.01% and its lowest is 86.52%, and for PTB-XL, fused Top-1 ranges from 85.03% to 89.64%, indicating high and stable re-identification.

4.3 Baseline and Comparison

Comparisons with prior ECG re-identification studies are not direct, as existing works typically evaluate models within the same population or allow for target adaptation. TRACE-ECG differs in its adversary knowledge and evaluation, as it enforces a strict cross-dataset transfer setting in which identity representations are learned on a source cohort and applied unchanged to fully disjoint external datasets, without target labels or fine-tuning. We provide a structured comparison in Table 3, reporting performance and key attack protocol. This comparison is not intended as a direct accuracy benchmark, but to highlight that despite the more constrained and realistic threat model, we achieve levels comparable to prior results obtained under less restrictive settings.

Table 3. Comparison of ECG Re-Identification Approaches and Evaluation Settings

Work	Datasets	Method	Protocol	Target Adap- tation?	Reported Result
Ghazarian et al. (2022) [16]	Single dataset (PhysioNet-scale)	Deep neural networks	Intra-dataset evaluation (same population)	Yes	~95-100% accuracy
Wang et al. (2024) [45]	Combined datasets (incl. MIT-BIH)	Feature-based + ML	Within-cohort evaluation	Yes	67-81.9% accuracy
Aguelal et al. (2025) [3]	Multiple public ECG datasets (incl. MIT-BIH, MIMIC-IV)	CNN-based + hand-crafted features fingerprints + similarity matching	Non-overlapping record splits within dataset	No target leakage, but same dataset adaptation	75-100% accuracy
Wang et al. (2025) [46]	Multiple public ECG datasets	DL-based representation + confidence thresholds	Partial-knowledge linkage within dataset	No target leakage, but same set adaptation	~85% accuracy
TRACE-ECG (ours)	ECG-ID, MIT-BIH, MIMIC-IV, PTB-XL	Frozen ResNet encoder + patient fingerprints + lead-wise fusion	Strict identity disjointness; encoder trained on population and deployed on external cohorts	No target labels, no fine-tuning, no target adaptation	85-96% (fused); and up to 100% for single-lead

Prior ECG re-identification studies report high accuracy under within-dataset or partially adaptive settings [16,45,3,46], confirming that ECG signals contain strong identity-discriminative patterns, yet in less constrained conditions and did not assess whether such representations generalize across independent populations. Despite the strict setting, our results show that re-identification remains highly effective even when the attack model is trained on a separate cohort and deployed unchanged on disjoint external datasets. Achieving performance comparable to prior work (75-100%), indicates that ECG identity information is not only learnable but also transferable across datasets, thereby representing a broader and more realistic privacy risk.

4.4 Confidence Analysis

To complement the attack results and quantify the reliability of an individual match, we perform a confidence analysis of linkage scores. For each leaked query, s_1 represents the fused top-1 score, and s_2 the fused score of the runner-up candidate, and define the margin $m = s_1 - s_2$. We then fit a logistic regression model that follows Platt scaling, used in literature [40,37]:

$$\hat{p}(\text{correct} \mid s_1, m) = \sigma(\alpha s_1 + \beta m + \gamma),$$

where $\sigma(\cdot)$ is the logistic sigmoid. For each source configuration, the parameters (α, β, γ) are estimated post hoc via logistic regression [19,27] using s_1 and m (capturing the separation between the top-ranked and competing candidates, a proxy for prediction confidence) as predictors and the correctness of the top match as the binary target. The logistic regression model is fitted post hoc using the observed closed-world evaluation outputs. It is therefore used to analyze score reliability and threshold rejection on the observed results. It is not used for model training, model selection, ranking, Top- k computation, or as an independently validated deployment threshold. This provides an empirical reliability score for analyzing how rejection thresholds behave across the evaluated matches. We therefore intend to validate linkage confidence in separating correct from incorrect matches, not to provide a fully conclusive open-set calibration model.

Correct cross-dataset matches generally receive higher post-hoc confidence scores. On the external targets, the mean post-hoc confidence score for correct matches is 0.929 in the MIMIC-source setting and 0.901 in the MIT-source setting, indicating that correct matches consistently receive higher confidence. Table 4 reports the number of correctly matched patients with empirical confidence above 0.80. For example, under the MIMIC-source configuration, 246 of 267 ECG-ID queries (92.13%), 133 of 141 MIT-BIH queries (94.33%), and 1,376 of 1,563 PTB-XL queries (88.04%) are correctly matched with confidence of at least 0.80. Under the MIT-source configuration, the corresponding counts are 237 of 267 ECG-ID queries (88.76%), 171 of 231 MIMIC-IV queries (74.03%), and 1,198 of 1,563 PTB-XL queries (76.65%).

Table 4. High-Confidence Re-identified Matches (≥ 0.80)

Source	Target	Queries	Correct matches ≥ 0.80	% of queries	True-match retention
MIMIC	ECG-ID	267	246	92.13%	98.80%
MIMIC	MIT-BIH	141	133	94.33%	99.25%
MIMIC	PTB-XL	1563	1376	88.04%	96.36%
MIT	ECG-ID	267	237	88.76%	98.34%
MIT	MIMIC-IV	231	171	74.03%	82.61%
MIT	PTB-XL	1563	1198	76.65%	87.19%

The last column shows that applying this threshold preserves most of the successful fused matches. For instance, under the MIMIC-source setting, 98.80% of the correct ECG-ID matches, 99.25% of the correct MIT-BIH matches, and

96.36% of the correct PTB-XL matches remain above the threshold. In other words, on the observed evaluation outputs, a threshold of 0.80 retains most successful matches while filtering out weaker predictions. This should be interpreted as an illustrative operating point rather than a validated deployment threshold.

While this is not a full open-world evaluation, it illustrates a practical accept/reject option for future open-set ECG linkage settings, where the queried identity may not always be present in the target gallery. In such cases, low-confidence predictions can simply be rejected by setting a minimum threshold, relative to the use case, rather than being forced to match the wrong patient. A full open-set calibration pipeline remains future work.

5 Mitigation Roadmap for ECG Data

Our results demonstrate stable biometric characteristics related to ECGs, enabling cross-dataset linkage. This reframes ECG privacy risks from a simple anonymization problem to a practical risk management challenge under different scenarios. In particular, it requires evaluating signal representation, accessibility, and models’ interactions with potential adversarial capabilities.

Previous studies [20] highlight the limitations of the single anonymization application, and we emphasize that privacy preservation must be adapted to the intended use of data and to the realistic adversarial capabilities, following a practical assessment instead of treating it as a generic privacy risk. In this section, we outline a roadmap for ECG data sharing that combines attack risk measurement with appropriate Privacy Enhancing Technologies (PET), such as DP.

5.1 ECG Re-identification Attack Surface

The attack presented in our work arises from the exposure of dimensions that explicitly influence the feasibility of the linkage.

Table 5. Exposure Dimensions on ECG Re-identification

Dimension	Description	Example Scenario
Data accessibility	Extent to which ECGs are publicly accessible	Research datasets or shared clinical repos
Signal representation	Level of detail preserved in the released signal	Raw ECG waveforms or high-resolution snippets
Model and transfer	Availability of model or embeddings derived from signals and learning	Foundation models or biometric encoders
Data components	Availability of features, aggregation, and amplifiers	Multi-lead aggregation that amplifies linkage

The attack pipeline evaluated in this work exploits mainly the points detailed in Table 5 through the assumption of publicly accessible datasets, detailed waveforms, and transferable representation learning.

5.2 Attack-Aligned and Risk-Adaptive Mitigation

Before applying privacy-preserving mechanisms, it is necessary to analyse the adversarial conditions under which ECG data may be linked. In our setting, the evaluation is performed through a cross-dataset linkage audit aligned with the threat model presented in Section 3.3. The linkage success obtained provides an operational measure of privacy risk. In practice, such an evaluation quantifies the ability of an attacker to recover patient identities based on our assumptions. Further, we underline the ECG-specific amplification factors (e.g., multiple leads) that strengthen identity matching. We view the attack as a readiness test before data release: data owners can use the assessment to identify the exposure points that enable linkage and select mitigation strategies accordingly.

This attack-aligned evaluation is the basis for deciding on any advanced or additional mitigation mechanisms that are necessary before data release, while maintaining relevant utility [11]. Furthermore, we illustrate a risk-adaptive roadmap in Figure 3 that links risk assessment with mitigation decisions. We define the scope of ECG release and audit for potential exposure sources (e.g., metadata, repeated patient recordings). A simulation aligned with the adversary model (including a potential roadmap for the presented attack) and then studied to estimate the re-identification risk.

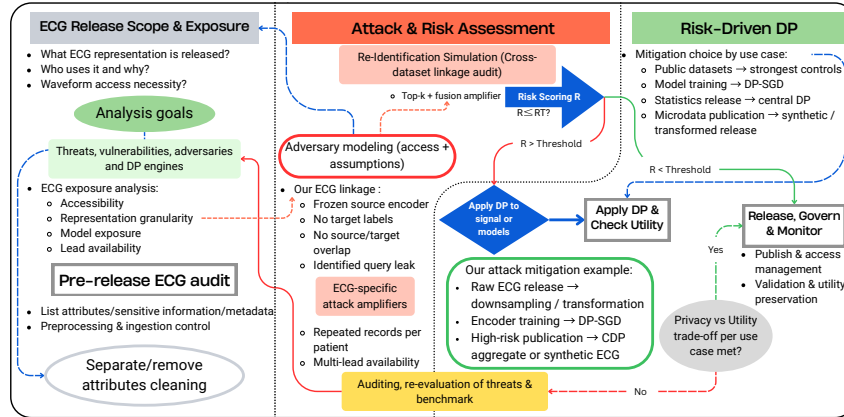


Fig. 3. Attack-aligned Risk Assessment & Mitigation Roadmap for ECG Data Sharing

The practical risk of ECG re-identification varies across use scenarios on how data are used in real-world systems. Including:

- **Clinical data sharing:** ECG datasets exchanged between hospitals or re-search groups.
- **Public dataset publication:** Open research datasets.
- **Foundation model training:** Large-scale models trained on aggregated ECG collections.

- **Wearable telemetry:** Continuous ECG acquisition from consumer health devices.

If the risk exceeds the acceptable level for the intended use case, mitigation mechanisms should be applied at different stages of the pipeline. Adapted mitigation mechanisms include transformation for mandatory raw ECG release, DP-SGD (Differentially private stochastic gradient descent) for model training/release [1] to prevent the learned embedding space from encoding strong identity signals, or alternative publication strategies such as synthetic data release [48].

5.3 Towards ECG Re-identification Evaluation

The roadmap presented Figure 3 outlines the importance of iterative evaluation when sharing physiological signals. Instead of relying on static anonymization, disjointed from practical attack assessments, protection can be guided by repeated adversarial testing and adjustments while verifying analytical value.

Future work involves extending this roadmap as an end-to-end framework for ECG data. We aim to deploy a systematic application combining linkage attack simulations benchmarking with suitable mitigation strategies across multiple data sharing scenarios while incorporating multi-seed training and confidence intervals to assess training variability beyond the current deterministic protocol.

6 Conclusion

Our work examines the privacy implications of electrocardiogram (ECG) data sharing through a strict cross-dataset linkage attack. We demonstrate that representations learned from a single source population can transfer to entirely disjoint external datasets and enable accurate linkage of de-identified ECG records, without identity overlap, target supervision or adaptation. Despite these strict adversarial constraints compared to prior within-dataset evaluations, the proposed attack achieves comparable and high accuracy on external datasets, including multi-lead fusion reaching between $\approx 85\%$ and 96% Top-1 accuracy. This indicates that embedded biometric characteristics could be exploited to support transferable matching attacks across independent populations, highlighting a broader and more realistic privacy risk. We also show that linkage risk is not only measurable at the aggregate level but can also be analyzed at the match level through post hoc confidence scoring. Although this confidence analysis is not a full open-world evaluation, it demonstrates that score margins and top-match scores can support rejection-based analysis, in which low-confidence matches are filtered rather than forced into an identity. This provides a practical direction for future open-set ECG linkage evaluation. ECG de-identification should not be considered a one-time, static process but rather part of a comprehensive, continuous risk management process. Evaluating de-anonymization risks through adversarial evaluation to quantify how released signals may be linked across

datasets. Furthermore, we introduce an attack-aligned roadmap that connects risk assessment with mitigation strategies. The presented approach emphasizes evaluating exposure sources, pre-study datasets, and use cases. In particular, transformations, privacy-preserving model training, and controlled publication strategies may contribute to reducing the de-anonymization likelihood while preserving analytical value.

In conclusion, the study underlines the high risk of ECG re-identification even under strict settings, and advocates incorporating realistic adversarial evaluations into health data sharing practices. Future research will focus on extending this evaluation into a larger benchmark framework that combines re-identification attacks in an open-world setting, benefiting from the confidence score we presented.

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